

THE INFLUENCE OF RESTING PERIOD LENGTH ON JUMPING PERFORMANCE

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ABSTRACT

Pereira, G, Almeida, AG, Rodacki, ALF, Ugrinowitsch, C, Fowler, NE, Kokubun, E. The influence of resting period length on jumping performance. *J Strength Cond Res* 22: 1259–1264, 2008—The purpose of this study was to determine a resting interval between countermovement jumps (i.e., volleyball spikes) that allows the maintenance of maximal jumping performance. Ten male volleyball players (1.85 ± 0.05 m, 77.2 ± 10.6 kg, 21.6 ± 5.3 years) performed 6 experimental jumping sessions. In the first and sixth sessions, maximal countermovement jump height was measured, followed by submaximal countermovement jumps to the point of volitional fatigue. The number of countermovement jumps was used as a reference to test the effect of rest period between volleyball spikes. From the second to fifth experimental sessions, 30 maximal volleyball spikes were performed with different resting periods (i.e., 8, 14, 17, and 20 seconds) followed by countermovement jumps. Between the 15th and 30th spikes, the blood lactate concentration and heart rate were measured. Because the performance on the first and sixth sessions was the same, no training effects were noticed. During the 8-second resting interval set, the lactate concentration increased significantly between the 15th and 30th spikes (i.e., from 3.37 ± 1.16 mmol to 4.94 ± 1.49 mmol); the number of countermovement jumps decreased significantly after spikes compared to those performed without a previous effort (i.e., from 23 ± 7 jumps to 17 ± 9 jumps); and these variables were significantly correlated ($r = -0.7$). On the other hand, the lactate concentration and number of countermovement jumps were stable across the other resting intervals, without a heart rate steady state. The results indicate that an adequate resting period between spikes allowed participants to achieve a lactate steady state in which the performance was maintained during the

exercise. These findings show that resting intervals between 14 and 17 seconds, typical during volleyball matches, are indicated to use in volleyball spike drills due to their capacity to maintain maximal jumping performance.

KEY WORDS blood lactate concentration, heart rate, intermittent, volleyball spike

INTRODUCTION

Sport performance depends on a number of factors. In some sports, one of these factors is the combination of the athlete's ability to generate and sustain a high work output. It has been known that such capabilities are antagonistic; the greater the work output is, the shorter the duration that can be sustained and vice versa. Thus, coaches should seek for training strategies to maintain maximal work output according to the activity needs. Depending on the sport characteristics, the control of both performance and physiological demand may help to identify the best training strategies.

In continuous exercise, when physiological metabolic equilibrium is attained, athletes can sustain a high work output for a long period (10). A steady state of both blood lactate concentration and heart rate is associated with physiological equilibrium (2,6). However, intermittent exercises can be sustained for longer periods than exercises performed continuously (1,8). This is possible only when the work-to-rest ratio allows a steady state to be attained (5,14).

If a steady state is not attained, depletion of substrates or accumulation of metabolites may produce performance loss in intermittent exercises (9,10). This effect can be partially or totally reversed by controlling the length of the resting interval (4,9). Thus, in sports in which movements are performed repetitively with a high muscle power output, the control of the resting time between exercises is an important variable to handle training intensity. Countermovement jumps are frequently trained under maximal conditions due to their importance in many sports. Volleyball players jump constantly during the game and the identification of an

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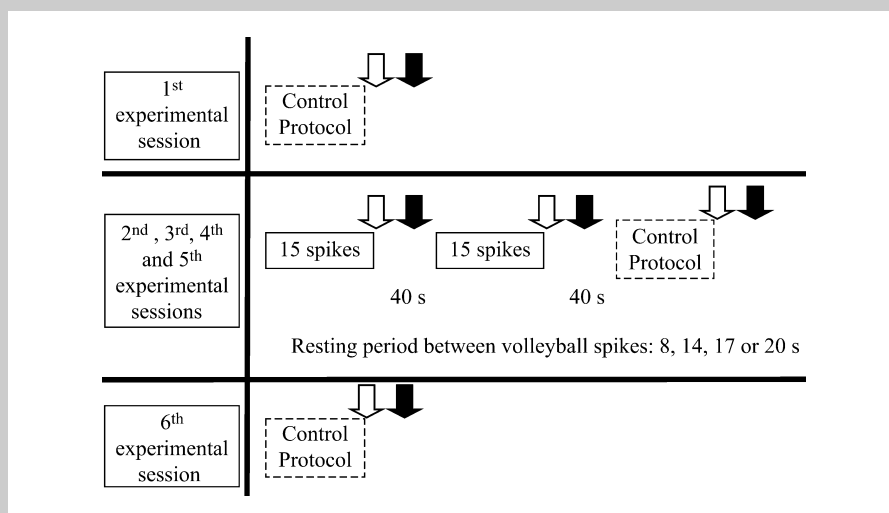


Figure 1. Schematic representation of the experimental design. Two different instants were represented when heart rate (empty arrows) and blood samples (filled arrows) were collected.

optimal resting interval between jumps can be of relevance for sport training.

Therefore, this study was performed to determine a resting period between countermovement jumps that allows the maintenance of maximal jump height. It was hypothesized that an adequate resting period between countermovement jumps would promote the maintenance of performance upon subsequent tasks. In addition, the high performance in

volleyball match (i.e., 8, 14, 17, and 20 seconds). Each experimental session had a fixed resting interval among jumps. Blood lactate concentration and heart rate were measured in the middle and at the end of the set (i.e., 15th and 30th spikes). It was hypothesized that there was an optimal resting interval that allows performance optimization (i.e., a physiological steady state) in a control condition (i.e., number of countermovement jumps) performed after the set

of 30 volleyball spikes. The control protocol determined the total number of jumps performed until fatigue.

METHODS

Experimental Approach to the Problem

A set of 30 volleyball spikes was performed in an intermittent mode to test the influence of resting interval length in the maintenance of maximal jump performance. In this set, subjects jumped as high as possible, and the length of the resting interval among jumps was altered. The length of resting intervals was based on typical intervals between rallies in a volleyball match (i.e., 8, 14, 17, and 20 seconds). Each experimental session had a fixed resting interval among jumps. Blood lactate concentration and heart rate were measured in the middle and at the end of the set (i.e., 15th and 30th spikes). It was hypothesized that there was an optimal resting interval that allows performance optimization (i.e., a physiological steady state) in a control condition (i.e., number of countermovement jumps) performed after the set

of 30 volleyball spikes. The control protocol determined the total number of jumps performed until fatigue.

Subjects

Ten healthy male volleyball players (1.85 ± 0.05 m, 77.2 ± 10.6 kg, 21.6 ± 5.3 years) volunteered for this study, trained twice a week for at least 5 years (mean, 5.4 ± 2.4 years), and participated in county competitions. Subjects were in the preseason during the experiment and were asked to maintain their habitual diet and physical activity level during the experiment. They were asked to refrain from severe physical activity on the day before the experimental session. This study was approved by the local ethical committee, and subjects were informed about

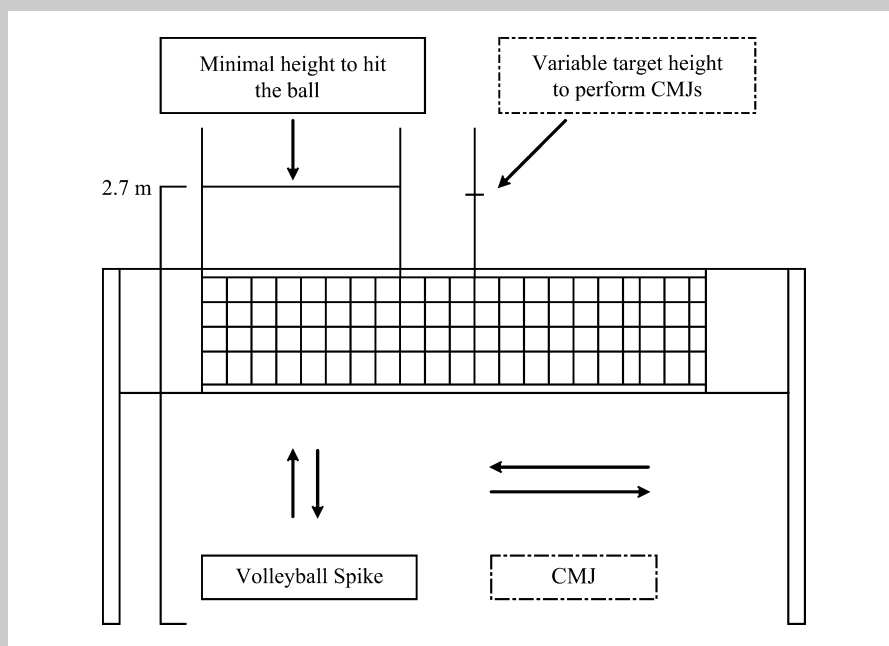
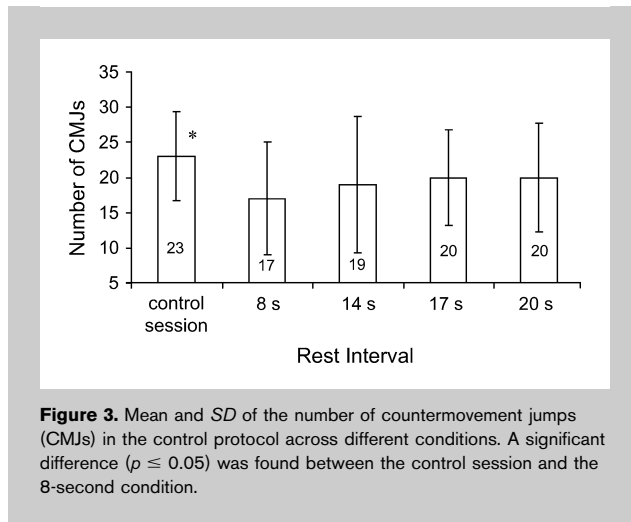


Figure 2. Schematic representation of the experimental setup. CMJ = countermovement jump.



the procedures and risks of the tests before giving written consent.

Experimental Design

Subjects performed 6 experimental sessions across 15 days. Experimental sessions were at least 2 days apart to avoid training and learning effects. A standardized warm-up was performed by players to ensure the same preparation level before testing (i.e., 5 minutes of running, 3 minutes of stretching, and 2 minutes of performing a few countermovement jumps). In the first experimental session, subjects performed a control protocol. In this situation, subjects had to perform volleyball spike movements by touching a fixed target 0.2 m below the maximal jumping height. They were considered fatigued when they no longer could touch the target. In the next 4 experimental sessions, subjects performed a set of 30 volleyball spikes with different resting intervals (i.e., 8, 14, 17, and 20 seconds). After each set, subjects performed the same control protocol as in the first experimental session. Finally, subjects repeated the control protocol without a previous

effort in the last experimental session to control for training effect. During all experimental sessions, subjects received verbal encouragement. Figures 1 and 2 show the experimental design and experimental setup, respectively.

First and Last Experimental Sessions

In the first session, the maximal reaching height mimicking volleyball spike movement was determined and recorded for the following sessions. This task was named countermovement jump to be different from 30 volleyball spikes performed in the other sessions. Movements were initiated from a distance of 2.5 m from the target, parallel to the volleyball net (Figure 2). The jumping technique was not controlled, but participants were recommended to perform similar countermovements. Subjects performed 5 trials with a 40-second resting interval. The best trial was considered the maximal jumping height (0.68 ± 0.06 m). The coefficient of variation among the 5 trials was $3.1 \pm 1.0\%$. Then, players performed countermovement jumps every 4 seconds to the point that they no longer (i.e., 3 consecutive times) reached a target set at 0.2 m below their maximal jumping height or returned to the initial position. This height corresponded to approximately 6% below their best performance. Submaximal jumping height was adopted based on a pilot study and on common variation found in this type of movement (13). The number of countermovement jumps was counted. At the end of the control protocol, blood samples were collected from the earlobe immediately after exhaustion to monitor lactate concentration with an electrochemical device (YSI 2300; YSI, Inc., Yellow Springs, OH). Heart rate was recorded every 5 seconds (Polar Sport Tester; Polar Electro Oy, Kempele, Finland) throughout the protocol and downloaded to a personal computer, from which the average of 3 heartbeats registered at the very end of the control protocol was calculated.

Second, Third, Fourth, and Fifth Sessions

Participants performed 30 volleyball spikes with intervals between each jump of 8, 14, 17, or 20 seconds. The resting interval order was randomly presented. Players were instructed to hit the ball as high as possible, and the minimum height allowed was 2.7 m above ground level (Figure 2). The ball was thrown by an experienced volleyball coach, who used a visual reference (i.e., a rope positioned in the same plane of the maximal ball height) to toss the ball consistently at the same height. A 40-second interval was introduced at the middle (i.e., after the 15th spike) and at the end (i.e., after the 30th spike) of the set to allow blood

TABLE 1. Mean $\pm$ SD blood lactate concentration (mmol) after the 15th and 30th volleyball spikes and after the control protocol in different conditions.

Condition	15th spike	30th spike	Control protocol
Control session	—	—	5.22 ± 1.05
8-second interval	$3.37 \pm 1.16^*$	$4.94 \pm 1.49^{\ddagger\S}$	$6.31 \pm 1.38^{\S}$
14-second interval	2.68 ± 1.37	$3.56 \pm 1.77^*$	$5.43 \pm 1.33^{\S}$
17-second interval	2.33 ± 0.89	2.62 ± 1.24	$5.18 \pm 1.00^{\S}$
20-second interval	1.94 ± 1.18	2.30 ± 1.41	$4.76 \pm 1.13^{\S}$

*Significantly different from the 20-second interval condition.

† Significantly different from the 17- and 20-second interval conditions.

‡ Significantly different from the 8-second condition after the 15th volleyball spike.

§ Significantly different from 15th and 30th volleyball spikes.

TABLE 2. Mean $\pm$ SD heart rate (b $\cdot$ min $^{-1}$) after the 15th and 30th volleyball spikes and after the control protocol in different experimental conditions.

Condition	15th spike	30th spike	Control protocol
Control session	—	—	186 $\pm$ 9*
8-second interval	175 $\pm$ 14†	182 $\pm$ 12‡	191 $\pm$ 9§
14-second interval	167 $\pm$ 16	176 $\pm$ 17‡	194 $\pm$ 9§
17-second interval	161 $\pm$ 16¶	167 $\pm$ 15‡	193 $\pm$ 7§
20-second interval	154 $\pm$ 15	162 $\pm$ 16‡	191 $\pm$ 7§

*Significantly different from the control protocol after the hitting set.

†Significantly different from the 14-, 17-, and 20-second interval conditions.

‡Significantly different from the 15th volleyball spike.

§Significantly different from the 15th and 30th volleyball spikes.

||Significantly different from the 17- and 20-second interval conditions.

¶Significantly different from the 20-second interval condition.

samples to be collected. The average of 3 heartbeats collected at the same instants (i.e., immediately after the 15th and 30th spikes) were used to represent the heart rate during the set. Forty seconds after the end of the hitting set, participants repeated the control protocol to determine the number of jumps that could be performed up to fatigue. The 8-second interval among volleyball spikes in the experimental session was based on findings of a pilot study, which showed that players were able to sustain performance throughout the 30-hit set. The intervals chosen among the other experimental conditions (i.e., 14, 17, and 20 seconds) were also selected with the aid of a pilot study, which revealed that steps shorter than 3 seconds caused important changes in the subsequent performance. These experimental conditions were also selected to represent a realistic game condition, similar to that reported in the literature (7,12).

Statistical Analyses

Power analysis revealed that the number of subjects for countermovement jumps, blood lactate concentration, and heart rate produced power values of 0.6, 1.0, and 1.0, respectively. A Kolmogorov-Smirnov test was applied and confirmed data normality. No order effect was seen on experimental sessions after testing for it. A 2-way ANOVA for repeated measures was applied to compare the mean number of countermovement jumps, blood lactate concentration, and heart rate in the control protocol and across experimental conditions (i.e., at the end of the control protocol performed in the first session, after the 15th and after the 30th volleyball spikes in the sessions with variable intervals of 8, 14, 17, and 20 seconds, and after the control protocol) and within experimental sessions (i.e., after the 15th and 30th volleyball spikes and after the control protocol). A Tukey test was used for post hoc comparison purposes. Linear regression was also applied to identify the relationship between the number of countermovement jumps in the control protocol and blood lactate

concentration in the experimental condition. Significance level was set at $p \leq 0.05$.

RESULTS

In the first and last control sessions, the number of countermovement jumps (21.2 ± 6.3 jumps and 25.2 ± 8.1 jumps, respectively), blood lactate concentration (5.37 ± 0.91 mmol and 4.84 ± 1.34 mmol, respectively), and heart rate (189 ± 11 b $\cdot$ min $^{-1}$ and 184 ± 9 b $\cdot$ min $^{-1}$, respectively) did not differ ($p > 0.05$). These results indicated no training effect and showed good reproducibility, as also suggested by the high intraclass

correlation coefficients (number of jumps, 0.82; blood lactate, 0.57; heart rate, 0.93). Therefore, the average between the first and last control sessions was considered for all analysis purposes.

The number of countermovement jumps performed in the control protocol after the 8-second resting interval condition decreased ($p \leq 0.05$) compared to that observed at the end of the control protocol without a previous effort. The other intervals (i.e., 14, 17, and 20 seconds) did not change the number of countermovement jumps in the control condition ($p > 0.05$). Figure 3 shows the variation in the number of countermovement jumps in the control session and in the control protocol performed after the experimental conditions.

Blood lactate concentration was significantly greater ($p \leq 0.05$) after the 30th than after the 15th spike in the 8-second condition only. In addition, lactate concentration was higher in the 30th spike of the 8-second session than in the 17- and 20-second sessions ($p \leq 0.05$) and in the 14-second session

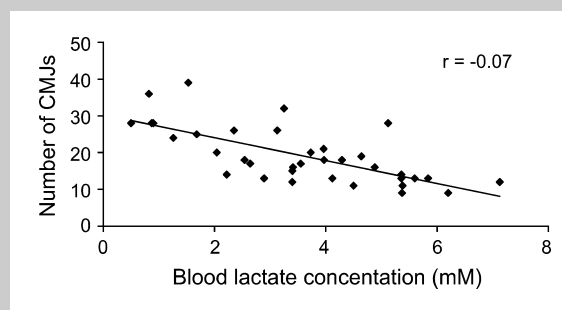


Figure 4. Pearson correlation coefficient between blood lactate concentration (mmol) after the 30th volleyball spike and the number of countermovement jumps (CMJs) in the control protocol after experimental conditions ($p \leq 0.05$).

than in the 20-second session. Blood lactate concentration was not different among the control protocols ($p > 0.05$) in all sessions. Table 1 summarizes the changes in blood lactate concentration in such conditions.

Heart rate in the control session was lower than in all experimental conditions. In the control protocol, it was greater than immediately after the 15th and 30th volleyball spikes in all experimental conditions, except in the control session, in which heart rates were considerably lower ($p \leq 0.05$). The heart rate between the 15th and 30th volleyball spikes increased significantly ($p \leq 0.05$) in all experimental conditions. Table 2 summarizes the heart rate changes in such conditions.

DISCUSSION

The main finding of this study was that an adequate resting period between countermovement jumps may promote a physiological steady state and therefore prolong work output on subsequent tasks (Figure 3). Resting intervals greater than 14 seconds neither change the number of jumps in the control protocol nor produce increments in blood lactate concentration increasing heart rate in a shallower pattern.

The experimental protocols were effective in controlling exercise intensity due to the increment in heart rate and blood lactate concentration, which was inversely proportional to the length of the resting interval. The unchanged blood lactate concentration and significantly smaller increment in heart rate, when the resting interval was longer than 14 seconds, suggested that a steady state was attained during the jumping task. This is in agreement with the high negative correlation found between blood lactate concentration and the number of countermovement jumps in this study (Figure 4) and in the literature (13).

A few studies found in the literature have manipulated the resting interval between the performances of explosive movements (e.g., volleyball spikes). Matsushigue (13) measured blood lactate concentration and heart rate in a similar experimental setting and showed a mean blood lactate concentration of 11.6 ± 1.3 mmol and 7.8 ± 1.6 mmol and mean heart rate of 185 ± 7 b·min⁻¹ and 176 ± 17 b·min⁻¹ for male volleyball players and nonplayers, respectively, after 40 volleyball spikes with an 8-second resting interval. The lower blood lactate concentration was observed in the current study under the same resting intervals (i.e., 57% and 37% lower, respectively). The differences between these 2 studies may be due to the small number of volleyball spikes performed in the current study. On the other hand, the heart rate was very similar (i.e., 2% and 3%) to that reported by Matsushigue (13).

Blood lactate concentration and heart rate have different kinetics (19). The former has an exponential growth after the anaerobic threshold has been reached, and the latter behaves in a logarithmic way and increases at a very shallow rate after the anaerobic threshold. Subjects performed 10 more spikes in the study by Matsushigue than in the current study. Thus, subjects in the previous study may have crossed the anaerobic threshold, resulting in an exponential growth of

blood lactate concentration. Subjects in the current study seemed to be close to the anaerobic threshold at the end of the 30-spike set. Nevertheless, heart rate has a very shallow rate of increment after the anaerobic threshold, which is the reason for the similar values between the current study and that by Matsushigue.

Chagas Neto et al. (7) reported that the number of spikes performed by male middle blockers and outside hitters in 5 matches was about 30 ± 8 jumps. This finding corroborates the idea that the number of jumps performed in the current study and in the real game was similar. It is also interesting to note that most intervals that occur between countermovement jumps during a volleyball match (i.e., 77% of a match) are similar to those in which a substantial change in a steady state of blood lactate concentration (i.e., between 14 and 20 seconds) was not found (12). This suggests that high fatigue levels may not be present during volleyball matches. In fact, a study performed by Rodacki et al. (16) showed that jump height after a game was unaffected, even when considering the number of jumps performed by each player. However, fatigue during a jump task may slow movement speed, although jumping height is not affected (11). Decreased movement speed may be critical for middle blockers, who need to jump high and fast to block quick sets.

Training sessions must be organized to consider such findings in order to avoid high levels of fatigue and changes in movement pattern to maintain performance. Bobbert and Van Soest (3) in a simulation study and Rodacki et al. (16) in an experimental study have shown that training stimulus specificity plays a fundamental role in maximal vertical jump performance. In fact, the adaptations and the effectiveness of an exercise program are closely related to the similarity between the training stimulus and the target skill (18). Thus, training programs that do not mimic the stimuli conditions (e.g., intensity and interval between exercises) may not change or even reduce jumping performance (3,18). This may be more important when athletes are performing specific volleyball drills.

Therefore, the current study supports the hypothesis that an adequate resting period among explosive exercises, particularly in countermovement jumps, may induce different exercise performance and physiological responses upon subsequent tasks. A resting interval from 14 to 17 seconds seems necessary to maintain performance in consecutive volleyball spikes during specific volleyball drills. A resting interval greater than 17 seconds could promote the same benefits as a lower rest interval (i.e., from 14 to 17 seconds), but the length of training sessions would be increased without necessity and the interval training efficiency could be decreased.

PRACTICAL APPLICATIONS

Volleyball drills are usually performed from several minutes to a few hours. Therefore, these drills should use an interval between spikes from 14 to 17 seconds in order to maintain jumping performance and movement pattern. This large

interval between spikes allows the achievement of a physiological steady state similar to that found in volleyball matches.

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